

**SECURITY
ASSESSMENT**

PREPARED FOR
CASH VAULT

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Revision History & Version Control

Version	Date	Author's	Description
1.0	April 21, 2026	Gurkirat, M Gowda	Initial Report

OxTeam conducted a comprehensive Security Audit to ensure the overall code quality, security, and correctness. The review focused on ensuring that the code functions as intended, identifying potential vulnerabilities, and safeguarding the integrity of operations against possible attacks.

1.0 Executive Summary

1.1 Overview

OxTeam partnered with the Cash Vault team to conduct a security audit of their smart contracts. The audit involved a manual review of the codebase to ensure the contracts are secure and reliable before deployment. This process was carried out to provide confidence to the project team and its community regarding the safety of the deployed contracts.



1.2 Scope

The scope of this audit involved a thorough analysis of the Cash Vault Smart contracts, focusing on evaluating its quality, rigorously assessing its security, and carefully verifying the correctness of the code to ensure it functions as intended without any vulnerabilities.

Language

SOLIDITY

Date of Received Scope

Apr 20, 2026

Initial Commit

NA

Fixed Commit

NA

Repository

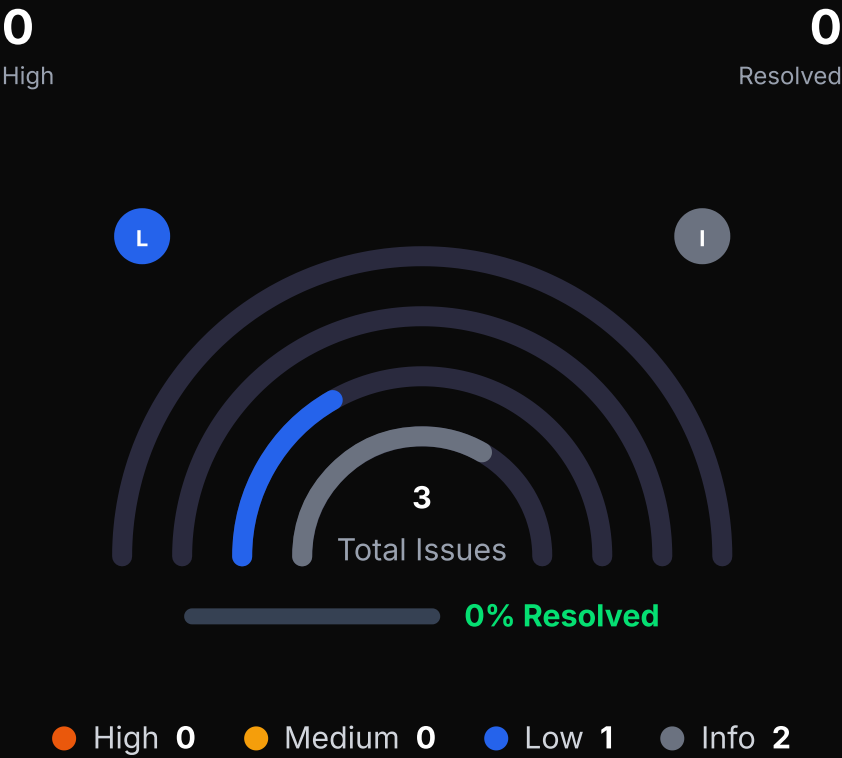
<https://basescan.org/address/0xe91A391be7fCE3C2fd5Feb97Aca4BbC854d8B63c#code#F1#L1>

● In-Scope Files

- contracts/CashVault.sol

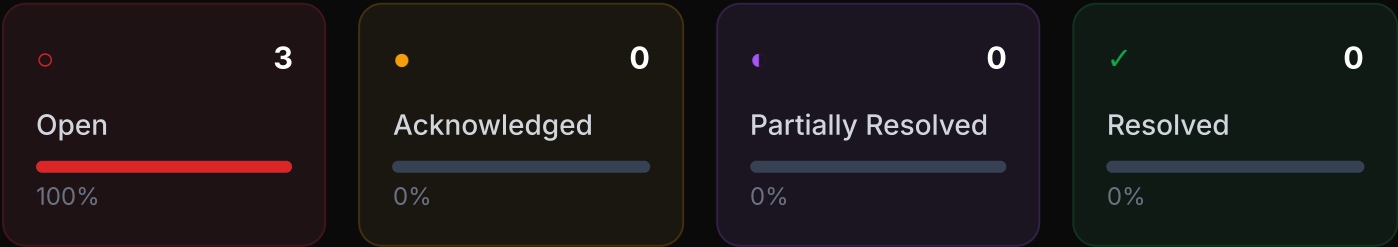
- **OUT-OF-SCOPE:** External Smart contracts code, other imported code.

1.3 Vulnerability Summary



1.4 Issue Status Summary

Status / Severity	High	Medium	Low	Info
Open	0	0	1	2
Acknowledged	0	0	0	0
Partially Resolved	0	0	0	0
Resolved	0	0	0	0



1.5 Type of Severity

High

Issues that pose an immediate and severe risk to the protocol's funds, user assets, or core system integrity, or significant security vulnerabilities that could lead to substantial financial loss or compromise of important protocol functionality. These issues require prompt attention and immediate remediation before deployment to prevent potential exploitation or complete loss of funds.

Medium

Moderate security concerns that could potentially be exploited under specific conditions or may impact protocol functionality. While not immediately critical, these issues should be resolved to ensure long-term security and reliability of the system.

Low

Minor issues that do not pose significant security risks but represent deviations from best practices or code quality standards. These findings improve code maintainability, readability, and adherence to industry conventions.

Informational

Observations and recommendations that enhance code quality, gas efficiency, or overall design. These are non-security findings that provide suggestions for optimization, documentation improvements, or architectural considerations.

1.6 Type of Issue Status

Open

Finding awaiting remediation by the development team.

Resolved

Issue has been addressed and verified by the audit team.

Acknowledged

Team has reviewed and accepted the risk without immediate fix.

Partially Resolved

Mitigation applied but complete resolution pending further work.

2. Techniques and Methods

The following techniques, methods, and tools were used to conduct a comprehensive security review of the smart contracts.

Structural Analysis

Examination of overall design, architecture, and code organization to ensure scalability, maintainability, and adherence to industry standards.

Static Analysis

Automated scanning for common vulnerabilities including reentrancy, arithmetic errors, access control issues, and potential denial-of-service vectors.

Manual Code Review

Line-by-line examination of contract logic to verify intended functionality, validate business requirements, and identify issues beyond automated detection.

Dynamic Analysis

Execution of contracts in controlled environments with comprehensive test cases to observe behavior, verify gas efficiency, and ensure correct state transitions.

Platforms and Tools Used to Audit

[Remix IDE](#)[Foundry](#)[Slither](#)[Hardhat](#)[Solhint](#)

3. Summary of Findings

ID	Title	Severity	Status
L-01	TOTAL_SUPPLY constant mints 100× the documented token supply	Low	Open
I-01	Centralisation Risk	Info	Open
I-02	Solidity Pragma Should Be Specific, Not Wide	Info	Open

● Resolved ● Partially Resolved ● Acknowledged ● Open

4. Findings

[L-1] TOTAL_SUPPLY constant mints 100× the documented token supply

LOW

Open

CONTRACT

CashVault

LOCATION

TOTAL_SUPPLY

Description

The contract defines the total token supply as:

```
uint256 public constant TOTAL_SUPPLY = 100_000_000_000 * 10 * 18;
```

This results in a total supply of **100 billion tokens**.

However, the contract documentation states:

"@notice Total fixed supply: 1 billion tokens"

Impact

This creates a **100× discrepancy between documented and actual tokenomics**:

- All allocations are **100× larger than documented**

This has several consequences:

- Misleading tokenomics for investors/users
- Incorrect assumptions in integrations (DEX listings, vesting, analytics)
- Potential financial damage

Recommendation

Ensure consistency between implementation and documentation.

Option A (Recommended — matches current documentation)

Update supply to 1 billion:

```
uint256 public constant TOTAL_SUPPLY = 1_000_000_000 10 * 18;
```

Option B (If 100B is intended)

Update all comments to reflect:

- 100 billion total supply
- Adjust all allocation numbers accordingly

[I-1] Centralisation Risk

INFO

Open

CONTRACT

CashVault

LOCATION

N/A

Description

The contract grants all the major roles to the admin, including:

- Executing the entire token distribution via `distributeTokens`
- Pausing and unpausing all token transfers via `pause()` / `unpause()`
- Minting tokens via `mint()`

Impact

By assigning roles to the same address, the system eliminates any operational separation and creates a **single point of failure**.

Protocols relying on RBAC must ensure:

- admin operations are restricted
- operational actions are delegated
- compromise of one role does not escalate to full system access

This design flaw removes these safety guarantees.

Recommendation

Assign Roles to Separate Addresses for admin, pauser, and minter.

[I-2] Solidity Pragma Should Be Specific, Not Wide

INFO

Open

CONTRACT

CashVault

LOCATION

N/A

Description

In the smart contract, the pragma directive `pragma solidity ^0.8.24;` is used to specify the compiler version. This directive allows any compiler version greater than or equal to 0.8.24;.

Impact

This directive allows any compiler version greater than or equal to 0.8.24;. Failure to specify a specific version may lead to compatibility issues or unexpected behaviour in future compiler versions. It's important to follow best practices to ensure the stability and security of the contracts

Recommendation

Update the pragma statements in the contracts to specify a particular version of Solidity. For example, replace `pragma solidity ^0.8.24;` with `pragma solidity 0.8.24;`.

```
&nbsp;pragma solidity 0.8.24;
```

Disclaimer

The OxTeam smart contract audit is not a security warranty, investment advice, or an endorsement of the Cash Vault Token. This audit does not guarantee that the code is completely free of vulnerabilities, as security risks can evolve over time. Users and investors should conduct their own due diligence before interacting with the Token. OxTeam is not responsible for any financial losses, hacks, or exploits that may occur after the audit. The findings in this report reflect the security state of the code at the time of assessment. By using this report, you acknowledge that smart contract security is an ongoing process and that OxTeam shall not be held liable for any direct or indirect damages resulting from the use of the audited code.

About OxTeam

OxTeam is a Web3-focused cybersecurity company dedicated to strengthening blockchain systems and decentralized applications. We help teams adopt blockchain technology with confidence by ensuring their smart contracts and infrastructure are secure, reliable, and production-ready.

Security is at the core of everything we do. Our services are designed to support smooth adoption across the Web3 ecosystem, enabling projects to scale safely while minimizing technical and financial risk.

OxTeam continues to grow its partnerships with developers, startups, and organizations across Web3, working together to build secure, resilient, and future-ready decentralized products.

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